

CO2 – AT2

Course Code: MLA0201

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Scenario – Based Questions

Q1) Apartment-Rent Prediction Using Linear Regression

Scenario Understanding

A real-estate company needs to predict apartment rent using floor area, number of rooms and location. Rent is a continuous numerical value, so this is a regression problem.

Key Factors

- Floor area in square feet
- Number of rooms
- Apartment location
- Historical monthly rent

Location must be converted into numerical dummy variables.

Application of Concept

A Multiple Linear Regression model can be used:

$$Rent = b_0 + b_1(Area) + b_2(Rooms) + b_3(Location)$$

The coefficients are estimated using historical apartment records. The dataset is divided into training and testing sets, and the model is evaluated using MAE, RMSE and R^2 .

Decision

Linear Regression is suitable because it predicts a continuous rent value and clearly shows how each feature influences rent. A larger area and better location are generally expected to increase the predicted rent.

Q2) Email Spam Classification Using a Linear Model

Scenario Understanding

A mail service must classify emails as spam or non-spam using the frequency of words such as “offer,” “free,” “win” and “prize.”

Key Factors

- Suspicious-word frequencies
- Number of links
- Message length
- Sender information
- Spam or non-spam label

Application of Concept

A linear classification model such as Logistic Regression can be used:

$$P(\text{Spam}) = \frac{1}{1 + e^{-z}}$$
$$z = b_0 + b_1X_1 + b_2X_2 + \cdots + b_nX_n$$

The email is classified as spam when the probability is at least 0.5.

Decision

Logistic Regression is appropriate because it produces both a class and a spam probability. Emails with high frequencies of suspicious words receive a higher spam score and can be automatically moved to the spam folder.

Q3) Flu Prediction Using Naive Bayes

Scenario Understanding

The medical system must predict whether a patient has flu using symptoms such as fever, cough and headache.

Key Factors

- Fever
- Cough
- Headache
- Prior probability of flu
- Symptom probabilities for flu and non-flu patients

Application of Concept

Naive Bayes calculates:

$$P(Flu | Symptoms) \propto P(Flu) \prod P(Symptom | Flu)$$

The same calculation is performed for the non-flu class. The patient is assigned to the class with the larger posterior probability.

Decision

Naive Bayes is suitable because the symptoms are categorical and the model works well with small medical datasets. However, the prediction should support a doctor and must not replace clinical diagnosis.

Q4) Loan Eligibility Using Discriminant Functions

Scenario Understanding

A bank must classify loan applicants as eligible or non-eligible based on income and credit score.

Key Factors

- Monthly or annual income
- Credit score
- Existing debt
- Previous loan-repayment history
- Eligibility class

Application of Concept

Linear Discriminant Analysis can create two discriminant functions:

$$\begin{aligned} g_1(X) &= w_1^T X + w_{10} \\ g_2(X) &= w_2^T X + w_{20} \end{aligned}$$

The applicant is assigned to the class with the larger discriminant value. The decision boundary is obtained from:

$$g_1(X) = g_2(X)$$

Decision

Applicants with sufficient income and a good credit score are more likely to be classified as eligible. Discriminant analysis is suitable when the classes can be separated approximately by a linear boundary.

Q5) Handwritten-Digit Recognition Using a Generative Model

Scenario Understanding

An optical character recognition system must classify handwritten images into digits from 0 to 9.

Key Factors

- Pixel intensity values
- Shape and stroke patterns
- Digit class from 0 to 9
- Prior probability of each digit
- Probability distribution of pixels within each class

Application of Concept

A probabilistic generative model such as Gaussian Naive Bayes models:

$$P(X | Digit) \text{ and } P(Digit)$$

It calculates:

$$P(Digit | X) \propto P(X | Digit)P(Digit)$$

The digit with the highest posterior probability is selected.

Decision

The generative model is simple and fast for basic OCR. For complex handwriting, convolutional neural networks may produce better accuracy, but Gaussian Naive Bayes clearly demonstrates probabilistic classification.

Q6) Customer-Churn Prediction Using a Discriminative Model

Scenario Understanding

A telecom company needs to predict whether a customer will leave its service using usage and account information.

Key Factors

- Monthly charges
- Customer tenure

- Call and data usage
- Number of complaints
- Contract type
- Churn label

Application of Concept

Logistic Regression is a probabilistic discriminative model:

$$P(\text{Churn} | X) = \frac{1}{1 + e^{-w^T X}}$$

It directly learns the boundary between churn and non-churn customers without modelling how the features were generated.

Decision

Customers with a churn probability above a chosen threshold are marked as high risk. The company can offer discounts or improved service plans to retain them. Recall is important because missing a likely churner may result in customer loss.

Q7) Product-Purchase Prediction Using Bayesian Logistic Regression

Scenario Understanding

An online advertising system must predict whether a customer will purchase a product after viewing an advertisement.

Key Factors

- Number of advertisement views
- Clicks on advertisements
- Time spent on the product page
- Previous purchases
- Customer age or interest
- Purchase outcome

Application of Concept

Bayesian Logistic Regression uses:

$$P(Y' = 1 | X\beta) = \frac{1}{1 + e^{-X\beta}}$$

A prior distribution is assigned to the coefficients:

$$\beta \sim N(0, \sigma^2)$$

The posterior combines prior knowledge with observed customer data.

Decision

A customer with a high posterior purchase probability should receive a targeted advertisement or offer. The Bayesian method is useful because it also represents uncertainty in the estimated coefficients.

Q8) Monthly-Sales Forecasting Using Linear Regression

Scenario Understanding

A retail company needs to forecast monthly sales using advertising expenditure and seasonal demand.

Key Factors

- Advertising expenditure
- Month or season
- Festival periods
- Previous sales
- Discounts and promotions
- Monthly sales value

Application of Concept

A Multiple Linear Regression model can be written as:

$$Sales = b_0 + b_1(Advertising) + b_2(Season) + b_3(Promotion)$$

Seasonal variables such as festival or non-festival months can be encoded as binary values.

Decision

The model estimates the expected monthly sales for a given advertising budget and season. The forecast supports inventory planning, staffing and marketing-budget allocation. RMSE and R^2 can be used to evaluate performance.

Q9) News-Article Classification Using Naive Bayes

Scenario Understanding

A news portal must categorize articles into sports, politics and entertainment based on their textual content.

Key Factors

- Article words
- Word frequency
- Headlines
- Category prior probabilities
- Sports, politics and entertainment labels

Application of Concept

Text is converted into count or TF-IDF features. Multinomial Naive Bayes calculates:

$$P(\text{Category} | \text{Words}) \propto P(\text{Category}) \prod P(\text{Word} | \text{Category})$$

The article is assigned to the category with the highest posterior probability.

Decision

Articles containing words such as “match” and “player” are more likely to be sports, while words such as “election” and “government” indicate politics. Naive Bayes is suitable because it is fast and effective for multiclass text classification.

Q10) Student-Placement Prediction Using Bayesian Logistic Regression

Scenario Understanding

A college placement system must predict whether a student will be placed using CGPA, coding skills and aptitude score.

Key Factors

- CGPA
- Coding-test score
- Aptitude score
- Communication skills
- Internship experience
- Placement outcome

Application of Concept

Bayesian Logistic Regression models:

$$P(P, laced \mid X\beta) = \frac{1}{1 + e^{-X\beta}}$$

Prior distributions are assigned to model coefficients, and placement data is used to calculate their posterior distributions.

Decision

Students with a high posterior placement probability are predicted as placed. Students with lower probabilities can be recommended for coding, aptitude or communication training. The model supports placement preparation but should not be used to permanently reject students.